

1. 10.52

$$A) H_0: p_{0.6} \leq p_{0.7} \quad H_0: p_{0.6} > p_{0.7}$$

$$TS = \frac{\frac{55}{70} - \frac{23}{70}}{\sqrt{\frac{78}{140} \left(1 - \frac{78}{140}\right) \left(\frac{1}{70} + \frac{1}{70}\right)}} = \frac{0.786 - 0.329}{\sqrt{(0.557)(0.443)\left(\frac{2}{70}\right)}} = 5.443$$

The R code below calculates the p-value for $z=5.443$

```
pnorm(5.443, lower.tail=FALSE)
## [1] 2.619532e-08
```

B) With a very small p-value ~ 0 , we reject the null hypothesis. We have very strong evidence that the true rate of normal RNA synthesis is lower for cells exposed to the higher concentrations of actinomycin D.

2. 10.63a

The following R code calculates the test statistic, p-value, and rejection region value for the data at $\alpha=0.05$.

```
yield <- c(785, 805, 790, 793, 802)
(mean(yield) - 800) / (sd(yield) / sqrt(5))
## [1] -1.341104
pt(-1.341104, 4, lower.tail=TRUE)
## [1] 0.1254872
qt(0.05, 4, lower.tail=TRUE)
## [1] -2.131847
```

With a large p-value of 0.1254872 and the test statistic -1.341104 is not within the rejection region $TS < -2.131847$, we fail to reject the null hypothesis. There is not enough evidence that the population average yield for the chemical process is less than 800 tons of chemical.

3. 10.98

$$A) L(y|\Theta_0) = \frac{1}{\Theta_0^n} m^n (\prod_{i=1}^n y_i)^{m-1} \exp(-\frac{\sum_{i=1}^n y_i^m}{\Theta_0})$$

$$L(y|\Theta_a) = \frac{1}{\Theta_a^n} m^n (\prod_{i=1}^n y_i)^{m-1} \exp(-\frac{\sum_{i=1}^n y_i^m}{\Theta_a})$$

$$\frac{L(y|\Theta_0)}{L(y|\Theta_a)} = \frac{\Theta_a^n}{\Theta_0^n} \exp \left[\left(\frac{1}{\Theta_0} - \frac{1}{\Theta_a} \right) (-\sum_{i=1}^n y_i^m) \right] < k$$

$$\exp \left[\left(\frac{1}{\Theta_0} - \frac{1}{\Theta_a} \right) (-\sum_{i=1}^n y_i^m) \right] < \frac{\Theta_0^n}{\Theta_a^n} k$$

$$\left(\frac{1}{\Theta_0} - \frac{1}{\Theta_a} \right) (-\sum_{i=1}^n y_i^m) < n \ln \frac{\Theta_0}{\Theta_a} + \ln k$$

$$(-\sum_{i=1}^n y_i^m) < (n \ln \frac{\Theta_0}{\Theta_a} + \ln k) \left(\frac{1}{\Theta_0} - \frac{1}{\Theta_a} \right)^{-1}$$

$$\sum_{i=1}^n y_i^m > -(n \ln \frac{\Theta_0}{\Theta_a} + \ln k) \left(\frac{1}{\Theta_0} - \frac{1}{\Theta_a} \right)^{-1}$$

$c = -(n \ln \frac{\Theta_0}{\Theta_a} + \ln k) \left(\frac{1}{\Theta_0} - \frac{1}{\Theta_a} \right)^{-1}$ so then $T = \sum_{i=1}^n Y_i^m > c$ is the rejection

region and c would be chosen to make the rejection region size α

Under the null $\frac{2T}{\Theta_0} = \frac{2 \sum_{i=1}^n Y_i^m}{\Theta_0} > \frac{2c}{\Theta_0}$ and follows a chi squared distribution

with $2n$ df. This does not rely on Θ_a , so it is UMP

B) Under the null, $T/50$ follows a chi squared distribution with $2n$ df. Then

$\frac{T}{50} > \chi_{0.05}^2$ is the rejection region and $\chi_{0.05}^2$ would be chosen so the test is size $\alpha=0.05$. Assuming the alternative, $T/200$ follows a chi squared distribution with $2n$ df.

$$\beta = P\left(\frac{T}{50} \leq \chi_{0.05}^2 | \Theta = 400\right) = P\left(\frac{T}{200} \leq \frac{1}{4} \chi_{0.05}^2 | \Theta = 400\right) =$$

$$P\left(\chi^2 \leq \frac{1}{4} \chi_{0.05}^2\right) = 0.05$$

$$\chi_{0.95}^2 = \frac{1}{4} \chi_{0.05}^2 \rightarrow 12=2n, n=6$$